FUNCTIONS AND COMPOSITIONS

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Every mathematical theory involves *objects* and *relations*, where relations describe how the objects interact with each other.

The objects we are currently working with are *sets* and the way we relate the sets are given by *functions*, that is, functions are a special class of relations on sets.

Now what are functions? They are simply ways we can relate two sets (not necessarily distinct) to each other! We can think of this as a (e.g. your computer) that takes in user inputs (pressing the SHIFT key and the "`q`" key) and spits out a single well-defined output (a upper case `Q`).

1. Functions as Relations Between Sets

Functions require three pieces of data:

(1) Set of inputs, which we call the *domain*
(2) Set of outputs, which we call the *codomain*
(3) A way of relating inputs with outputs (this is how the function is defined, usually with a *rule* such as $f(x) = 3x^2 + x$ or something similar you've seen in Algebra 2).

These pieces of data are subject to two conditions, which we call *axioms*:

(1) Defined on Domain: Every input has a well-defined output under the function rule
(2) Vertical Line Test: Given two different representations of the same input, our output does *not* change.

Notation. Let A and B be sets. A *function* f with *domain* A and *codomain* B is denoted with $f : A \to B$.

Let's look at some examples of the *failures* of condition (1) and (2), respectively.

Failure of Defined on Domain: Suppose I say

My statement

Let $f : \mathbb{R} \longrightarrow \mathbb{R}$ be a *function* defined by the rule $f(x) = 1/x$.

The incorrect part of my statement is that we cannot say the domain of f is $\mathbb{R}$ since $0 \in \mathbb{R}$ and we know that $f(0) = 1/0$ is undefined.

The correct thing to say would have been "let $f : \mathbb{R} \setminus \{0\} \longrightarrow \mathbb{R}$ be a function defined by the rule $f(x) = 1/x$ since the domain is all real numbers, except zero.

For the failure of Vertical Line Test: If $f : A \to B$ is a function, and I have (a, b) and (a, b') where $a \in A$, then we can conclude that $b = b'$.[1]

For the failure of (2), we can look at vertical lines of the graph of f to determine whether or not f is a function. If you are given a table of (x, y) values however (that table is a relation on sets!), the vertical line test condition says that if there is any repetitions in the x place found in the table, the y values should match otherwise that relation is *not* a function.

Example 1. Let $S = \{1, 2, 3\}$ and $T = \{0, 1\}$. Define a function $f : S \to T$ by letting $f(1) = 0, f(2) = f(3) = 1$.

(1) What is the domain of f?
(2) What is the codomain of f?
(3) Write down the set of all (x, y)-values you get under $f : S \to T$

(See footnote[2] for solution.)

The *range* (or *image*) of a function $f : A \to B$ is the subset of B

$$\mathrm{Im}(f) \stackrel{\text{def}}{=} \left\{ f(a) \in B : a \in A \right\}.$$

[1] If you are confused by what I mean by "different representation", this is a technicality that is taken care of by the above sentence (see me if you're really curious about this).

[2] Domain is $S = \{1, 2, 3\}$, the codomain is $T = \{0, 1\}$ and set of (x, y)-values is $\{(1, 0), (2, 1), (3, 1)\}$.

Range vs Codomain

The codomain is a set I (the teacher) have to give you; it is the set of all *possible* outputs your function might give while the range is the set of outputs that you *do* get. An example is if we think of the codomain as a bucket and the range is all of the objects that go inside the bucket.

Example 2.

(a) Let $A = \{x, y, z\}$ and $B = \{1, 2, 3\}$. Define a function $f : A \to B$ by $f(x) = f(y) = 1$, and $f(z) = 2$:

$$A \longrightarrow B$$

$$\begin{tikzcd}
x \arrow[r] & 1 \\
y \arrow[ur] & 2 \\
z \arrow[ur] & 3
\end{tikzcd}$$

and so we have

$$\begin{aligned}
\text{domain of } f: &\quad A = \{x, y, z\} \\
\text{codomain of } f: &\quad B = \{1, 2, 3\} \\
\text{image/range of } f: &\quad \{1, 2\}
\end{aligned}$$

(b) Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = 3$. Then we have

$$\begin{aligned}
\text{domain of } f: &\quad \mathbb{R} \\
\text{codomain of } f: &\quad \mathbb{R} \\
\text{image/range of } f: &\quad \{3\}
\end{aligned}$$

(c) Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = 2x^2$. Then we have

$$\begin{aligned}
\text{domain of } f: &\quad \mathbb{R} \\
\text{codomain of } f: &\quad \mathbb{R} \\
\text{image/range of } f: &\quad [0, \infty)
\end{aligned}$$

(d) Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = 9x$. Then we have

$$\begin{aligned}
\text{domain of } f: &\quad \mathbb{R} \\
\text{codomain of } f: &\quad \mathbb{R} \\
\text{image/range of } f: &\quad \mathbb{R}
\end{aligned}$$

(e) Let $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ be defined by $f(x) = 9/x$. Then we have

$$\begin{aligned}
\text{domain of } f: &\quad \mathbb{R} \setminus \{0\} \\
\text{codomain of } f: &\quad \mathbb{R}
\end{aligned}$$

$$\text{image/range of } f: \quad \mathbb{R} \setminus \{0\}$$

1.1. **Types of Functions.** Let A and B be sets and let $f : A \to B$ be a function. Then

(1) f is *injective* or *one-to-one* if for all $a, a' \in A$:

$$f(a) = f(a') \Longrightarrow a = a'.$$

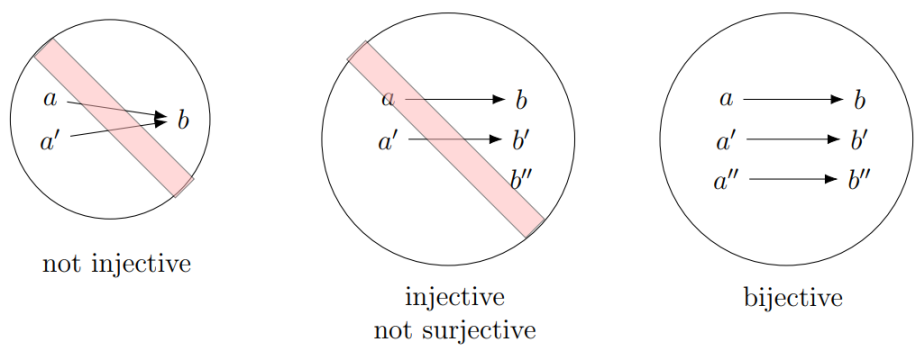
That is, a function is injective *if* every output (in range) has *exactly one* input.

(2) f is *surjective* or *onto* if

$$\mathrm{Im}(f) = B.$$

That is, f is surjective if its range fills up the entire codomain.

(3) f is *bijective* if it is both injective and surjective (one-to-one and onto).



Example 3. A function that is *not* injective is any function that fails the *horizontal line test*, e.g., $f(x) = x^2$. If we wanted to treat $f(x) = x^2$ as if f was injective, then by definition of injective functions, we would have

$$f(-3) = 9 = f(3) \Longrightarrow -3 = 3$$

which we know is false and so we conclude that $f(x) = x^2$ is not injective.

Example 4. A function that is injective is any function that passes the horizontal line test! If we are given a table instead, we just have to confirm that every output in our range is uniquely determined by a single input (no two inputs go to the same output).

Consider $f(x) = x^3 + 1$. This is injective since we can check that if we have an $x, y \in \mathbb{R}$ such that $f(x) = f(y)$, then:

$$\begin{aligned} f(x) = f(y) &\Longrightarrow x^3 + 1 = y^3 + 1 && \text{definition of } f \\ &\Longrightarrow x^3 = y^3 \\ &\Longrightarrow x = y && \text{taking cube roots of both sides} \end{aligned}$$

and so we conclude that our inputs must have been the same thing.

1.2. **Cardinality (Optional reading).** Given a set A, we define the *cardinality* of A to be the number of objects inside of the set A. The cardinality is denoted with absolute value bars around the set: $|A|$.

For example, $A = \{☕, 2, Y\}$ has cardinality $|A| = 3$ since there are three objects. The cardinality of $\mathbb{Z}$ is $|\mathbb{Z}| = \infty$ (there are an infinite number of integers).

We can now give an ordering on sets dependent on their size (cardinality), using functions.

Let A and B be sets. We say that

(1) $|A| \leq |B|$ if there exists an injective function $f : A \to B$ (there just has to be at least one injective function to exist!).
(2) We say that $|A| \geq |B|$ if there exists a surjective function $f : A \to B$ (again, there just has to be at least one surjective function to exist!).
(3) We say that $|A| = |B|$ if there exists a bijection $f : A \to B$.

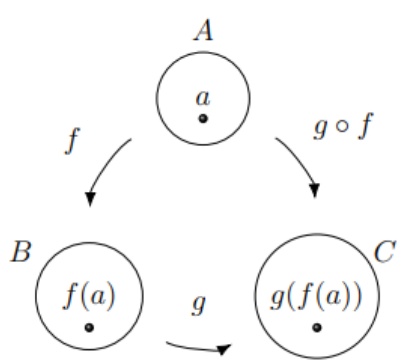
In this class we will eventually show that $|\mathbb{Z}| < |\mathbb{R}|$ (there is no surjection $f : \mathbb{Z} \to \mathbb{R}$ or an injection $g : \mathbb{R} \to \mathbb{Z}$); this gives us that there are different types of infinities and some are larger than others![3]

2. Compositions

Let A, B, C be sets and let $f : A \to B$ and $g : B \to C$ be functions. The *composition of f and g* is the (new) function

$$g \circ f : A \longrightarrow C$$

defined by $(g \circ f)(a) \overset{\text{def}}{=} g\left(f(a)\right)$. Pictorally, we have



where the diagram is saying if we start our life at $a \in A$, then it does not matter if (1) we take the f train to $f(a)$, and then the g train taking us to $g(f(a))$, or (2) take the express

[3]Don't think too hard about this; it's just one of the wacky sounding things out of mathematics. If you're interested about different infinities and how there's an unsolved problem about them, you may want to explore the *Continuum hypothesis*.

$g \circ f$ train and end up at $(g \circ f)(a)$. The reason why it does not matter is because we end up at the same destination since we defined $(g \circ f)(a) = g(f(a))$ for all $a \in A$.

Example 1. Let $f(x) = 3x + 1$ and $g(x) = \sin(11x) - 4$. Compute $(g \circ f)(x)$ and $(f \circ g)(x)$.

We first will compute $(g \circ f)(x)$ and then $(f \circ g)(x)$. See that

$$\begin{aligned}
(g \circ f)(x) &= g\big(f(x)\big) && \text{by definition of composition} \\
&= g\,(3x + 11) && \text{since } f(x) = 3x + 1 \\
&= \sin\big(11\,(3x + 1)\big) - 4 \\
&= \sin(33x + 11) - 4. \\
(f \circ g)(x) &= f\big(g(x)\big) && \text{by definition of composition} \\
&= f\big(\sin(11x) - 4\big) && \text{since } g(x) = \sin(11x) - 4 \\
&= 3\big(\sin(11x) - 4\big) + 1 \\
&= 3\sin(11x) - 12 + 1 \\
&= 3\sin(11x) - 11.
\end{aligned}$$

- when dealing with compositions, treat the inner function as an entire variable
- If asked to solve for either the inner or the outer function when given the composition (and one of the functions), then again try to pattern match as much as possible.

The second tip above probably made no sense, so let's look at examples:

Example 2. (Unknown inner) Let $f(x) = x - 3$. Find a $g(x)$ such that $(f \circ g)(x) = 2x - 1$.

Again, treat the $g(x)$ as an entire variable, and since $g(x)$ is the inner function, we plug that into $f(x)$ to get

$$2x - 1 = (f \circ g)(x) = f\big(g(x)\big) = g(x) - 3.$$

Thus, we have $g(x) - 3 = 2x - 1 \implies g(x) = 2x + 2$.

Example 3. (Unknown outer) Let $(g \circ f)(x) = x^2 + 6x + 10$ with $f(x) = (x + 3)$. Find what $g(x)$ is.

This is a little more complicated, but if we are careful and remember our algebra skills, we can identify every instance of $f(x)$ hidden in the composition:

$$\begin{aligned}
(g \circ f)(x) = x^2 + 6x + 10 &= x^2 + 6x + 9 + 1 \\
&= (x + 3)(x + 3) + 1 \\
&= \big(f(x)\big)^2 + 1
\end{aligned}$$

so you can conclude that $g(x) = x^2 + 1$.

3. Binary Operations

Every operation in math you have known in your life so far has been some form of *binary operation*, i.e., do some operation on two objects. For example, $+, -, \times, \div$ are all binary operations on $\mathbb{R}$.

We now formally define a binary operation: Let A be a set. A *binary operation* $\star$ *on* A is a function $\star : A \times A \longrightarrow A$.

What the previous definition means is: let $a, b \in A$. Then the function $\star(a, b)$ gives an output that is also in A. For binary operations, we often write $a \star b$ instead of $\star(a, b)$. We also have special names for certain properties about binary operations:

If $\star$ is a binary operation on A and we say that $\star$ is

(1) *associative* if for any $a, b, c, \in A$, we have
$$\big(a \star (b \star c)\big) = \big((a \star b) \star c\big)$$
(2) *commutative* if for any $a, b \in A$, we have
$$a \star b = b \star a.$$

Moreover, if there exists an element $\mathcal{O}_\star \in A$ such that for every object $a \in A$, we have
$$\mathcal{O}_\star \star a = a = a \star \mathcal{O}_\star,$$
then we say that $\mathcal{O}_\star$ is the $\star$-*identity element* of A.

Given a fixed object $a \in A$, if there exists another object $b \in A$ such that
$$a \star b = \mathcal{O}_\star = b \star a$$
then we say that b is the $\star$-*inverse of* a and b is (usually) denoted as a^{-1}.

Example 1. Let $A = \mathbb{Z}$ and let $\star$ be addition. We know that if $a, b \in \mathbb{Z}$, then so their sum $a + b \in \mathbb{Z}$ as well (the sum of any two integers is an integer) and therefore we know for a fact that $+$ is a binary operation on $\mathbb{Z}$.

We learn in elementary/middle school that $+$ is associative; you can add numbers in any order you want and you also learned that $+$ is commutative e.g. $2 + 5 = 5 + 2$ and you know that this property holds for every integer.

You've also known that adding 0 to any number does nothing; $0 = \mathcal{O}_+$ in this case and so 0 is the *additive identity* on $\mathbb{Z}$ since $a + 0 = a = 0 + a$.

You know that every integer has an *additive inverse*, namely its negative, e.g., the additive inverse of 5 is -5 since $5 + (-5) = 0 = (-5) + 5$ where 0 is the additive identity.

We note that the notation $5^{-1} = -5$ under addition *is correct*. However, when dealing with the $+$ operation, the inverses are denoted simply as subtractions (to avoid confusion with multiplication operations).

Example 2. Let $A = \mathbb{Z}$ and let $\star$ be multiplication. We know that if $a, b \in \mathbb{Z}$, then their product $a \times b \in \mathbb{Z}$ as well and so $\star = \times$ is a well defined binary operation on $\mathbb{Z}$.

We know that we can multiply integers in any order we want, i.e., if $a, b, c \in \mathbb{Z}$, then

$$a \times (b \times c) = (a \times b) \times c.$$

Moreover, we know that $ab = ba$ when we are dealing with integers, e.g., $4 \times 3 = 3 \times 4$.

Now we know that multiplying by 1 does nothing. That is, if $a \in \mathbb{Z}$ we have

$$a \times 1 = a = 1 \times a$$

and so $1 = \mathcal{O}_\times$, i.e., 1 is the *multiplicative identity* on $\mathbb{Z}$.

We note that multiplicative inverses do not necessarily exist in $\mathbb{Z}$. For example, $2 \in \mathbb{Z}$ and there does not exist an integer $m \in \mathbb{Z}$ such that

$$2m = 1$$

Example 3. Let $A = \mathbb{Q}$ and consider $\div$. Since if $a, b \in \mathbb{Q}$, then $a/b \in \mathbb{Q}$ gives us that $\div$ is a binary operation on $\mathbb{Q}$.

$\div$ is not associative, since

$$(2 \div 3) \div 4 = \frac{\frac{2}{3}}{4} = \frac{1}{6}$$

while

$$2 \div (3 \div 4) = \frac{2}{3/4} = \frac{8}{3}.$$

$\div$ is not commutative either, since $2 \div 3 = 2/3$ while $3 \div 2 = 3/2$.

Anything divided by 1 returns itself, i.e., $a \in \mathbb{Q}$, $a \div 1 = a$ but we note that $1 \div a \neq a$ unless $a = 1$. That is, there is no division-identity on $\mathbb{Q}$, i.e., $\mathcal{O}_\div$ does not exist in $\mathbb{Q}$ (it has to work both ways, not just one way).

Example 4. A really strange example: Let A be a set. Now let $\mathcal{P}(A)$ denote the set whose objects are *all subsets of* A.

A quick example would be: $A = \{1, 2\}$ then $\mathcal{P}(A) = \left\{\emptyset, \{1\}, \{2\}, \{1, 2\}\right\}$.

Now note that the union of any two subsets of A is still a subset of A and similarly, since the intersection of two subsets of A is a subset of A, we can define binary operations $\cup$ and $\cap$ on $\mathcal{P}(A)$! (not on A itself since functions require the individual inputs to be objects of the set, not the entire set itself).

Is $\star = \cup$ associative? i.e, if $S, T, U \subseteq A$, is $(S \cup T) \cup U = S \cup (T \cup U)$?

Is $\cup$ commutative? i.e., if $S, T \subseteq A$, is $S \cup T = T \cup S$?

Does there exist an $\cup$-identity element? Does there exist a $\mathcal{O}_\cup \subseteq A$ where if you take $S \subseteq A$, we have

$$S \cup \mathcal{O}_\cup = S = \mathcal{O}_\cup \cup S?$$

Do $\cup$-inverses exist for every $S \subseteq A$?

Now do the exact same for $\star = \cap$.

Similarly, we can introduce binary operations to function spaces. Let A, B be sets and also suppose addition, subtraction, multiplication, and division are well-defined binary operations on B.[4] Let $f, g : A \to B$ are functions, and then define

$$\begin{aligned} f + g &: A \longrightarrow B & f - g &: A \longrightarrow B \\ (f+g)(x) &= f(x) + g(x) & (f-g)(x) &= f(x) - g(x) \\ fg &: A \longrightarrow B & \frac{f}{g} &: A \longrightarrow B \\ (fg)(x) &= f(x) \cdot g(x) & \left(\frac{f}{g}\right)(x) &= \frac{f(x)}{g(x)} \end{aligned}$$

4. Identity and Inverses

Let A be a set. Every set defines at least one function....given any set A (even if $A = \emptyset$), we can define a function

$$\begin{aligned} \mathrm{id}_A &: A \longrightarrow A \\ \mathrm{id}_A(a) &= a \end{aligned}$$

This function looks oddly a lot like the line $y = x$ and that is because the line

$$\begin{aligned} \ell &: \mathbb{R} \longrightarrow \mathbb{R} \\ \ell(x) &= x \end{aligned}$$

is the identity map on functions $f : \mathbb{R} \to \mathbb{R}$ *under function composition*!

That is, if we let $A = \{f : \mathbb{R} \to \mathbb{R}\}$ be the set of all functions (whose domain is $\mathbb{R}$ and codomain is $\mathbb{R}$), and if $f, g \in A$, then the composition is a function $(g \circ f) : \mathbb{R} \to \mathbb{R}$ and so $(g \circ f) \in A$ as well; this shows that $\circ$ is a binary operation on functions.

Identity and Inverse Elements. In every example we've seen, we searched for the existence of an *identity element*, an object $\mathcal{O}$ in the set that leaves any object in the set unchanged when combined with $\mathcal{O}$. which leaves any element of the set unchanged when combined with it.

[4] we only need this extra condition because the adding/subtracting/whatever action is going on is happening *inside the codomain*. We aren't really doing any action with the domain and whatever action is happening in the codomain must be well-defined.

Now if there does exist an identity element $\mathcal{O}$, then given any object x in your set, can we find another object y in the set such that $x \star y = \mathcal{O} = y \star x$? This is the role of *inverses*: to get you back to the identity element.

Apply this idea to functions: Let $f : A \to B$ be a given function and let $g : B \to A$ be some function, but we want to make g into the inverse of f.

Applying the definition of an inverse, we **want** g to be the function such that

$$\begin{aligned} g \circ f : A &\longrightarrow A \qquad \text{and} \qquad & f \circ g : B &\longrightarrow B \\ (g \circ f)(a) &= \mathrm{id}_A(a) = a & (f \circ g)(b) &= \mathrm{id}_B(b) = b \end{aligned}$$

Here, the first equation (the one on the left) is saying that g must undo everything that f does. The second equation (the one on the right), is saying that f will undo everything that g does.

This discussion is to highlight this key point that causes confusion:

> Let $f : \mathbb{R} \to \mathbb{R}$ be a function with an inverse $f^{-1} : \mathbb{R} \to \mathbb{R}$. Then
> $$f^{-1}(x) \neq \frac{1}{f(x)}.$$
> Instead, f^{-1} is the function such that
> $$(f \circ f^{-1})(x) = x \quad \text{and} \quad (f^{-1} \circ f)(x) = x.$$

The reason is because functions are relations, not numbers. It is okay to say that $2^{-1} = \dfrac{1}{2}$ since *the binary operation* for this is *multiplication* and the identity element for $\mathbb{R}$ under multiplication is 1.

The identity element for functions, under composition, is the function $\ell(x) = x$ and since our binary operation on functions is different from the binary operation (multiplication) on numbers, we have different "formulas" for inverses.

Given a fixed function $f(x)$, in order for us to discuss a function $g = f^{-1}$ whose role is to "undo" everything f does, we need f to be a *bijection*![5]

5. Computing Inverses and Evaluation at Points

Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{6, 7, 8, 9, 10\}$. Suppose $f : A \to B$ is an invertible function with the following values

[5] See me for details about this; it comes down to the ensuring g satisfies the axioms of a function as defined at the beginning of the writeup.

inputs x	outputs $f(x)$
1	10
2	8
3	7
4	6
5	9

We should be familiar with evaluating $f(x)$, e.g., $f(1) = 10$ and $f(2) = 8$ and $f(3) = 7$ and $f(4) = 6$ and $f(5) = 9$.

By the definition of the inverse, we need a function $g : B \to A$ such that

x	$f(x)$	$(g \circ f)(x)$
1	10	1
2	8	2
3	7	3
4	6	4
5	9	5

That is, (after rearranging the rows of the table above) we have

inputs x	outputs $g(x)$
6	4
7	3
8	2
9	5
10	1

So $f^{-1}(6) = g(6) = 4$ and we check that

$$(f \circ g)(6) = f\big(g(6)\big) = f(4) = 6$$

and we see that the compositions output matches our input.

Now compare the two table of function values below:

x	$f(x)$
1	10
2	8
3	7
4	6
5	9

and

x	$g(x)$
10	1
8	2
7	3
6	4
9	5

What do you notice?

5.1. **The Switch Trick.** In the previous example/table, you should have noticed that the columns switch and that gives us the inverse function at specified points.

That is, if we ended up swapping the x and y's (where $y = f(x)$ that is).

This is generally true; if $f(x)$ is a function, we *might* be fortunate enough to be able to find a formula for its inverse $f^{-1}(x)$.

The method is given a function $f(x)$, set $y = f(x)$ and then replace all of the x variables with y. The final step is then to solve for y. The result is your function $f^{-1}(x)$.

Example 1. Let $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^3 + 1$ (see example 4). We will compute the inverse for this function:

To find $f^{-1}(x)$, first set $y = f(x)$ and replace all of the x variables with y:

$$\begin{aligned} y = x^3 + 1 &\rightsquigarrow x = y^3 + 1 && \text{swap } x \text{ and } y\text{'s} \\ &\rightsquigarrow y^3 = x - 1 && \text{subtract 1 from both sides} \\ &\rightsquigarrow y = \sqrt[3]{x-1} && \text{taking the cube root to isolate } y \end{aligned}$$

and so we have found that $f^{-1}(x) = \sqrt[3]{x-1}$ is the inverse of $f(x) = x^3 + 1$.

However, an explicit formula (a nice looking formula just as in example 1) for $f^{-1}(x)$ in general is not always possible; we can still talk about the values the inverse might take though! Consider the example (there is a similar problem on the homework)

Example 2. Let $f : \mathbb{R} \to \mathbb{R}$ be an invertible function given by $f(x) = x^3 + x^2 - 1$. Find the value of $f^{-1}(11)$.

Instead of trying to use the switch trick, simply remind yourself to use the definition of the inverse, and so this question is equivalent to asking "what is the value of $x \in \mathbb{R}$ such that

$$x^3 + x^2 - 1 = 11?$$

We can solve this! Simplify that $x^3 + x^2 - 1 = 11$ and ask yourself what x must be equal to.

$$12 = x^3 + x^2 = x^2 (x + 1) \Longrightarrow x = 2$$

since $2^2(2+1) = 4(3) = 12$. For this example, you have to take educated guesses (an *ansatz*) and check that it works.

This method of computation is necessary since it is incredibly difficult to compute a formula for $f^{-1}(x)$ for this example (so we can only evaluate the inverse at certain points).